

AI in Healthcare – Data Preparation and Analysis

Data Preparation, Summarization, Visualization, Reduction and Diagnostic Performance Analysis of a Prostate Cancer Cohort

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AI in Healthcare Summer Camp – Data Preparation, Summarization, Visualization and Reduction

Abstract

Three raw datasets (demographic/clinical, biochemical, comorbidity codes) were cleaned, merged and imputed into a single table (1,159 patients, 25 variables). PSA was evaluated as a metastasis marker via ROC (AUC=0.670). PCA reduced ten biochemical markers to three components (69.3% variance). A logistic regression model with backward elimination and 10-fold cross-validation reached AUC 0.855 (accuracy 0.783), outperforming PSA alone. SVM and random forest performed similarly. LIME explained one patient's prediction. PSA alone is a moderate discriminator; combining biomarkers, comorbidities and demographics improves prediction substantially.

Keywords: prostate cancer, PSA, ROC, PCA, logistic regression, cross-validation, LIME

Contents

1 Introduction

2 Data

3 Data Preparation and Results

3.1	Type correction and reshaping	1
3.2	Missing data and imputation	2
3.3	Merging	2
3.4	Patient characteristics	2
3.5	Correlation structure	2
3.6	PSA diagnostic performance	2
3.7	Principal Component Analysis (PCA)	3
3.8	Univariate logistic regression (urinary retention)	3
3.9	Multiple logistic regression and collinearity . .	3
3.10	Backward elimination	3
3.11	Cross-validated performance	3
3.12	Model comparison	3
3.13	LIME explanation	4

4 Conclusion

5 Annexe

1. Introduction

Prostate cancer (PC) is common and diagnosis relies on heterogeneous data: clinical records, laboratory tests, and medical history. Before modelling, data must be cleaned and merged. This report covers data preparation, visualisation, feature reduction and classification. The main questions: (i) how to prepare three separate sources into one reliable table, and (ii) how well does PSA alone discriminate metastasis, and can the biomarker panel be compressed without losing too much information.

2. Data

Three source files, linked by patient ID:

- **DATASET1** – clinical: civil status, family cancer history, metastasis (outcome), dates (diagnosis, metastasis, birth).
- **DATASET2** – laboratory: ten numeric markers (monocyte, urea, lymphocyte, thrombocyte, erythrocyte, albumin, creatinine, globulin, PSA, coagulation factor).
- **DATASET3** – medical history: one row per condition (diabetes, hearing loss, hypercholesterolaemia, hypertension, osteoporosis, urinary retention).

3. Data Preparation and Results

3.1. Type correction and reshaping

Method: Converted categorical variables in DATASET1 to proper types; parsed dates. DATASET2 types were already correct. DATASET3 was pivoted from long to wide format, creating one binary column per disease.

Result: All variables now in correct format for analysis.

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3.2. Missing data and imputation

Method: Checked missingness. DATASET1 had 63.7% missing in metastasis date (expected for non-metastatic patients). DATASET2 had scattered missingness (0.3–14.1%). Histograms (Fig. 1) showed most markers skewed, so we used median imputation (robust to outliers).

Result: Missing data handled; no patient loss.

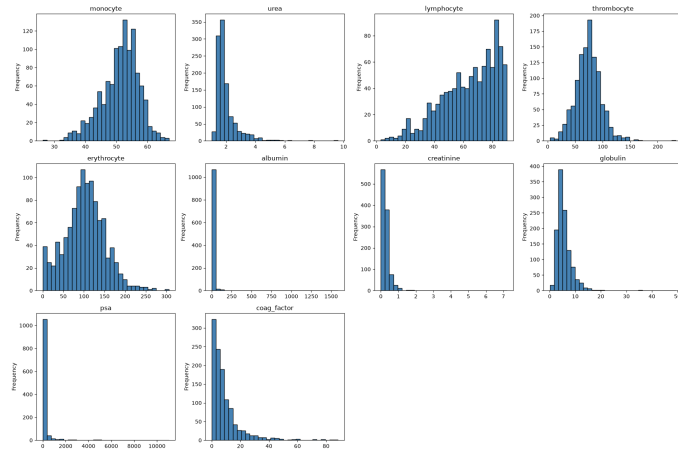


Figure 1: Distributions of ten biochemical markers.

3.5. Correlation structure

Method: Computed Pearson correlations among the ten biomarkers and visualised as heatmap (Fig. 2).

Result: Most correlations weak; no strong collinearity. All markers can be used.

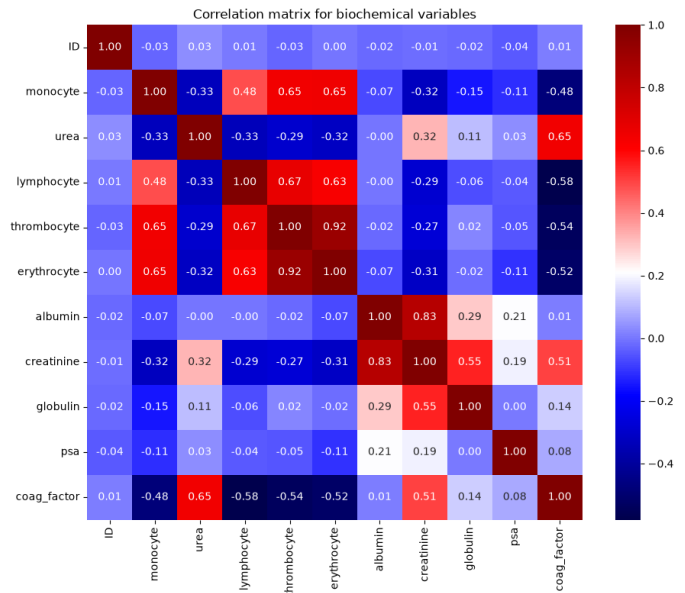


Figure 2: Correlation matrix of biochemical markers.

3.3. Merging

Method: Left-joined DATASET1, DATASET2 and the wide disease table on ID. Missing disease flags set to 0.

Result: Final merged dataset DATASET123: 1,159 rows $\times$ 25 columns.

3.4. Patient characteristics

Table 1 summarises the cohort.

Table 1: Patient characteristics (N=1,159).

Variable	Value
Age at PC diagnosis (yr)	74.3 $\pm$ 8.7
Age at metastasis (yr)*	78.9 $\pm$ 8.5
Metastasis: yes	421 (36.3%)
Family history: yes	623 (53.8%)
Civil status: married/cohabiting	808 (69.7%)

*only for patients with metastasis.

About one third developed metastasis, on average 4.6 years after diagnosis.

3.6. PSA diagnostic performance

Method: Tested two fixed cutoffs (20 and 100 ng/mL) for metastasis classification; performed ROC analysis and determined optimal cutoffs via Youden index and maximum accuracy.

Result: Table 2 shows performance. Lower cutoff gives higher sensitivity; higher cutoff gives higher specificity. Accuracy similar (0.65) due to class imbalance.

Table 2: PSA performance at fixed cutoffs.

	PSA >20	PSA >100
Sensitivity	0.596	0.323
Specificity	0.672	0.867
Accuracy	0.645	0.670

ROC curve (Fig. 3) gave AUC = 0.670 – moderate discrimination. Youden index chose 23 ng/mL; max accuracy chose 134 ng/mL.

ROC curve - PSA as predictor of metastasis

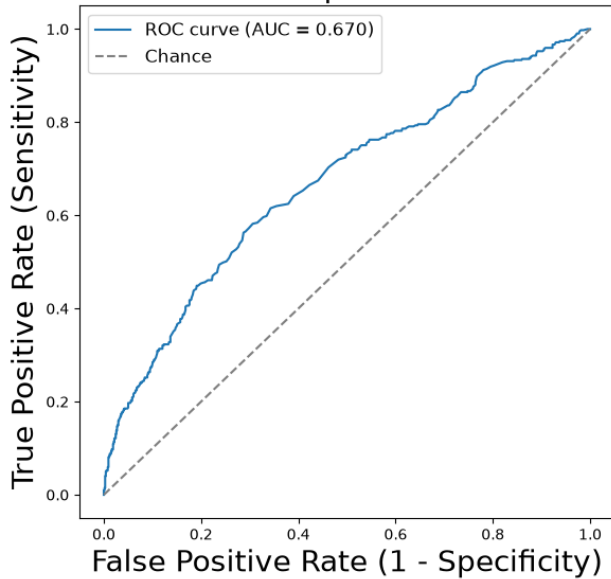


Figure 3: ROC curve for PSA (AUC=0.670).

3.7. Principal Component Analysis (PCA)

Method: Standardised biomarkers and performed PCA. Retained components with eigenvalue > 1 (Kaiser criterion).

Result: Three components retained (eigenvalues 3.90, 1.95, 1.10), explaining 69.3% of variance (Fig. 4). Loadings in Table 3. PC1: cellular markers, PC2: renal/nutritional, PC3: PSA-dominated. Good dimensionality reduction.

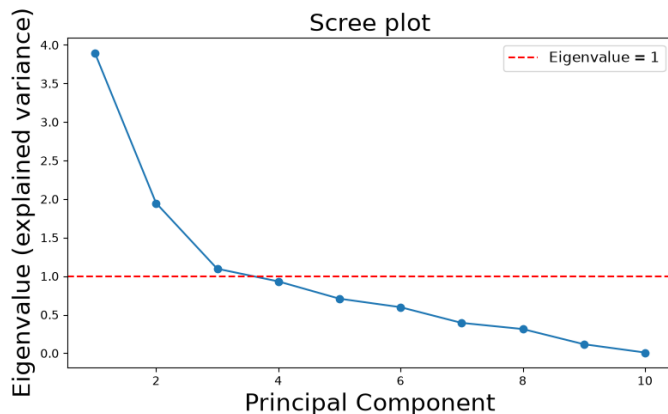


Figure 4: Scree plot with Kaiser cutoff.

Table 3: PCA loadings (top per component).

Marker	PC1	PC2	PC3
monocyte	0.381	0.112	-0.137
urea	-0.290	-0.041	-0.479
lymphocyte	0.348	0.167	0.051
thrombocyte	0.412	0.245	-0.225
erythrocyte	0.412	0.199	-0.260
albumin	-0.132	0.593	0.248
creatinine	-0.323	0.532	-0.047
globulin	-0.128	0.436	-0.345
psa	-0.085	0.173	0.600
coag_factor	-0.405	-0.063	-0.296

3.8. Univariate logistic regression (urinary retention)

Method: Fitted simple logistic regression: metastasis ~ urinary_retention.

Result: Significant (OR=2.49, $p < 0.001$) but low $R^2 = 0.029$. Accuracy at 0.5 cutoff: 65%, recall 37%.

3.9. Multiple logistic regression and collinearity

Method: Full model with all 21 predictors (excluding ID, dates, PCA components). Checked VIF.

Result: Pseudo $R^2 = 0.320$. High VIF for creatinine, albumin, coagulation factor (Table 4) – collinearity issue. Replacing biomarkers with PC1-3 solved VIF.

Table 4: VIF values (raw biomarkers).

Variable	VIF
Creatinine	63.4
Albumin	35.4
Coagulation factor	14.7
Globulin	6.6
Thrombocyte	5.3

3.10. Backward elimination

Method: Removed predictors one by one based on p-value until all remaining significant ($\alpha = 0.05$).

Result: Dropped 8 variables (including creatinine and coagulation factor). Final model had 13 predictors, AIC improved. In-sample performance at 0.5 cutoff: accuracy 0.79, recall 0.68, precision 0.72 (Table 5).

Table 5: Final model in-sample performance.

Class	Precision	Recall	F1
No metastasis	0.82	0.85	0.84
Metastasis	0.72	0.68	0.70
Accuracy	0.79		

3.11. Cross-validated performance

Method: 10-fold stratified cross-validation on final model.

Result: Mean metrics: accuracy 0.783 (SD 0.028), sensitivity 0.675 (SD 0.058), specificity 0.846, precision 0.718, F1 0.693, AUC 0.855 (SD 0.023). Stable. Huge improvement over PSA alone (AUC +0.185).

3.12. Model comparison

Method: Compared logistic regression, SVM, random forest, L1-logistic via same CV.

Result: All performed similarly (Table 6). Logistic regression has interpretable coefficients, so preferred.

Table 6: Cross-validated performance by model.

Model	Acc.	Sens.	Prec.	F1	AUC
Logistic Reg.	0.783	0.675	0.718	0.693	0.855
SVM	0.786	0.713	0.705	0.707	0.841
Random Forest	0.785	0.682	0.716	0.696	0.843
Reg. LR (L1)	0.783	0.677	0.717	0.694	0.855

3.13. LIME explanation

Method: Used LIME on one random patient to explain prediction.

Result: Top influential features: family history, albumin, diabetes, age, lymphocytes. LIME shows direction of each contribution for that patient.

4. Conclusion

Interpretation: We produced a clean merged dataset. PSA alone is a moderate discriminator (AUC 0.670), no single cut-off is perfect. PCA reduces ten markers to three components retaining 69% variance. A multivariable logistic model (13 predictors, cross-validated AUC 0.855) beats PSA significantly and remains interpretable. SVM and random forest give similar performance but are less transparent.

Impact: The logistic model can support risk stratification. The ROC analysis provides data-driven cutoffs when only PSA is available.

Limitations: Retrospective, cross-sectional design limits causal claims. Issues: median imputation should be verified; PSA>20 flag duplicates continuous PSA; diabetes coefficient negative (unexpected) – needs expert review. Future prospective studies with time-to-event data could use survival analysis.

5. Annexe

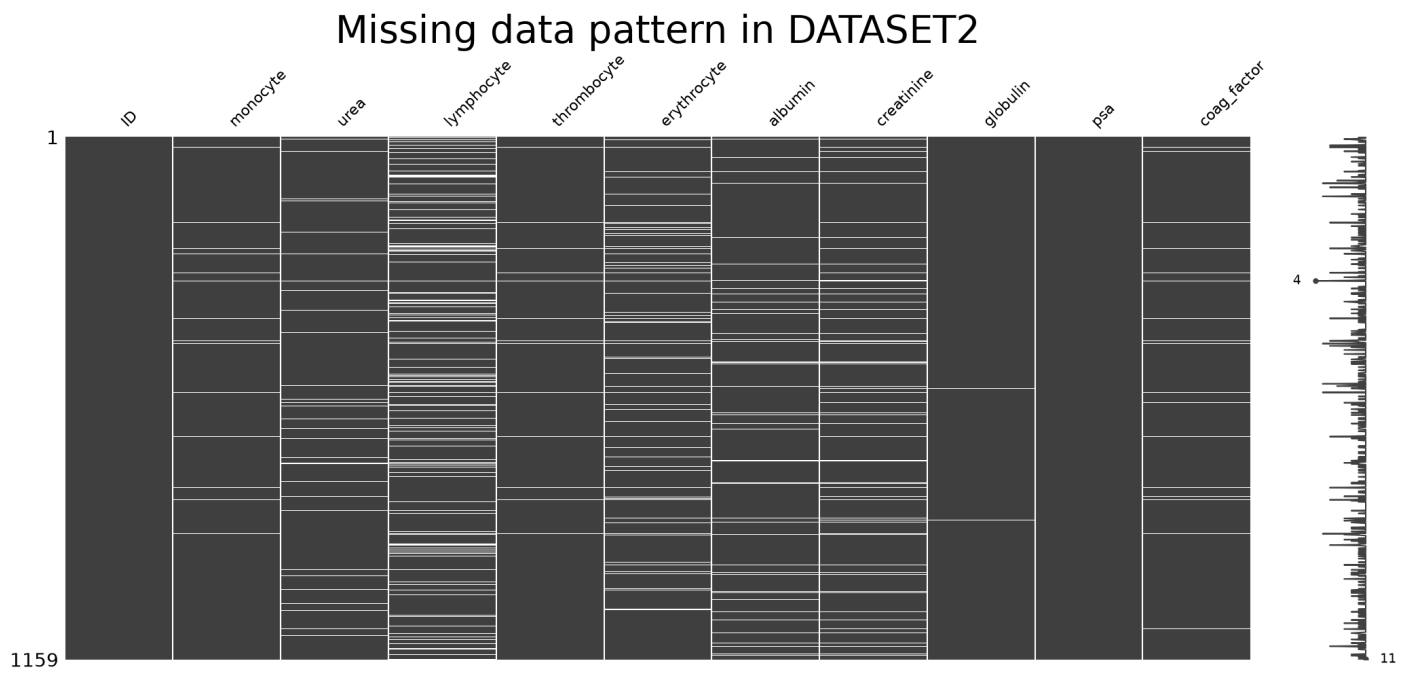


Figure 5: Missingness pattern in DATASET2 (scattered).